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SciScore Report

Below you will find your SciScore report containing three tables. Your score is calculated based on adherence to scientific rigor criteria (Table 1) and identification of key biological resources (Table 2). Table 3 contains statistical tests and oligonucleotides but is not scored. If SciScore makes any mistakes, please contact us to help us learn and improve.

Table 1: Rigor Adherence Table

Ethics
not detected.
Inclusion and Exclusion Criteria
not required.
Attrition
not required.
Sex as a biological variable
not detected.
Subject Demographics
Age: not detected.
Weight: not detected.
Randomization
For each positive PSM in which the peptide included two or more Ser/Thr/Tyrs , we kept the retention time and the ion intensity unchanged and moved the correctly localized phosphate group to a randomly chosen wrong localization to create the M_Sim1_LocF group .
Blinding
not detected.
Power Analysis
not detected.
Replication
not required.
Cell Line Authentication

not detected.	
Code Information	
Identifiers: Experimental Procedures Data Sources Preprepared data for AutoRT base model training and testing were downloaded from the GitHub website (https://github.com/bzhanglab/AutoRT/tree/master/example/data).	https://github.com/bzhanglab/AutoRT/tree/master/example/data
Data Information	
Identifiers: The MaxQuant search results of the phosphoproteomic data from HeLa cell protein extract used to train and test AutoRT base_phospho model were downloaded from PRIDE (https://www.ebi.ac.uk/pride/) with accession number PXD015087.	PXD015087
Identifiers: The MaxQuant search results of the synthetic dataset used to evaluate Delta RT and spectral similarity were downloaded from PRIDE (https://www.ebi.ac.uk/pride/) with accession number PXD000138, which included 96 peptide libraries generated from 96 seed peptides.	PXD000138
Identifiers: The raw files of an MS3-based TMT10-labeled phosphoproteomic dataset from murine cell lines (mouse_TMT) were downloaded from PRIDE (https://www.ebi.ac.uk/pride/) with accession number PXD015284.	PXD015284
Identifiers: The raw files of a label-free quantification phosphoproteomic dataset from human cell cultures (human_LFQ) were downloaded from PRIDE (https://www.ebi.ac.uk/pride/) with accession number PXD007145.	PXD007145
Identifiers: RT Data Preparation and AutoRT AutoRT base model was trained on unmodified peptides from the PXD006109 dataset (37 , 49) .	PXD006109

Table 2: Key Resources Table

Your Sentences	REAGENT or RESOURCE	SOURCE	IDENTIFIER
Experimental Models: Cell Lines			
The MaxQuant search results of the phosphoproteomic data from HeLa cell protein extract used to train and test AutoRT base_phospho model were downloaded from PRIDE (https://www.ebi.ac.uk/pride/) with accession number PXD015087.	HeLa		Suggestion: RRID:CVCL_0030(link)
Experimental Models: Organisms/Strains			
In our study, we used only a subset of data including 11 raw files (03275_A4_P038189_U09 - 03275_H3_P038189_U08), which were the phosphopeptides analyzed by a nano-flow LC-MS/MS system.	03275_A4_P038189_U09 - 03275_H3_P038189_U08		
One experiment with six fractions was downloaded (02277_A01_P024190_S00_U01_R1 - 02277_F01_P024190_S00_U06_R1).	02277_A01_P024190_S00_U01_R1 - 02277_F01_P024190_S00_U06_R1		
Recombinant DNA			
For pDeep2 base_phospho model , there were 7205 PSMs for training and 2506 PSMs for testing .	pDeep2		
Software and Algorithms			
The raw files of a label-free quantification phosphoproteomic dataset from human cell cultures (human_LFQ) were downloaded from PRIDE (https://www.ebi.ac.uk/pride/) with accession number PXD007145.	PRIDE		Suggestion: Pride-asap (RRID:SCR_012052)(link)
The second one is MS-GF+ v2017.01.27 combined with a localization tool PhosphoRS, which we referred to as CDAP.	MS-GF+		Suggestion: MS-GF+ (RRID:SCR_015646)(link)
56,431 Peptide Identification and Phosphosite Localization For phosphoproteomic data used to train and test AutoRT base_phospho model , search results were downloaded from PRIDE (https:// www.ebi.ac.uk/pride/) with accession number PXD015087 (MaxQuant v1.6.2.352	https:// www.ebi.ac.uk/ pride/		
UniProtKB Human Reference Proteome database v22.07.13) .	Human Reference Proteome		

Besides , all the three benchmarking datasets were processed by FragPipe pipeline.	FragPipe		
We followed the data preparation rules of AutoRT: when a peptide has multiple PSMs , which means this peptide has multiple RTs , only the peptide with the RT range smaller than 3 min would be kept , and there should be no peptide sequence overlaps between training and test data to avoid overfitting problem .	AutoRT		
General Phosphorylation Probability We provided RefSeq human and UniProtKB mouse reference peptide libraries described above as input for the MusiteDeep server (https:// www.musite.net) , which returned the general phosphosite probabilities of serine , threonine , and tyrosine in the peptide sequences (44) .	RefSeq		
	UniProtKB		
Experimental Design and Statistical Rationale For UCEC, global phosphoproteomics data including 16 experiments were processed by MaxQuant as one batch.	MaxQuant		Suggestion: MaxQuant (RRID:SCR_014485)(link)
Data analysis was performed using Python language (Jupyter Notebook v 5.6.0, Python 3).	Python		Suggestion: python-bx (RRID:SCR_024202)(link)

Other Entities Detected

Your Sentences	Recognized Entity
Statistical Tests	
We considered PCC , Spearman's Correlation Coefficients (SPC) , cosine similarity (cos) , Kendal rank correlation coefficient (kdt) , and spectral angle (SA) as possible measurements of spectral similarity , and they led to similar conclusions .	Spearman's Correlation
	Kendal rank correlation
Two-sided Wilcoxon rank sum test was performed, and p-values were corrected with Bonferroni correction.	Wilcoxon rank sum test
	Bonferroni

SciScore is an automated tool that is designed to assist expert reviewers by finding and presenting formulaic information scattered throughout a paper in a standard, easy to digest format. ***SciScore is not a substitute for expert review.*** SciScore also checks for the presence and correctness of several unique identifiers, including RRIDs (research resource identifiers) in the manuscript, detects sentences that appear to be missing RRIDs, and can even suggest RRIDs under certain circumstances. **All RRID suggestions should be verified;** only the author can know whether the suggestions are correct.

For a full description of scored criteria and tips for improving your score, please see https://www.scicrunch.com/sciscorereport-faq